Sahand Rahmdel

Education

M.Sc. Computer Science, Specialization: Artificial Intelligence and Data — Aachen, Germany
RWTH Aachen University — *Apr. 2025 – Est. Mar. 2027*

- *Relevant Coursework:* Machine Learning, Advanced ML, Data Science, Reinforcement Learning, Computer Vision, Business Process Intelligence, Actions & Planning in AI, Quantum Computing
- *Seminar Neuroinspired Computing – Reinforcement Learning* **[Link]**: A paper on computational framework and biological foundations of Reinforcement Learning; Grade 1.0.

B.Sc. Computer Science including Coursework in Business Administration — Aachen, Germany
RWTH Aachen University — *Nov. 2020 – Mar. 2025*

- *Seminar Neuroinspired Computing – Attention Mechanism and its Applications* **[Link]**: A paper on biological vs. computational Attention Mechanism; presented a deep-dive on the Transformer; Grade 1.3.

Work Experience

AI Engineer (Working Student) — Nov. 2025 – Present
Mercedes-Benz AG — *Sindelfingen, Germany - Remote*

- Sole engineer and technical lead contact of NEXA (NX Engineering eXpert Agent), an enterprise AI agentic companion; presented live architecture to 800+ attendees (500 in-person) at company town hall.
- Architected NEXA's multi-agent LangGraph pipeline with local RAG (semantic and lexical search tools with reranking) and custom MCP services to automate code generation across a 70k+ LOC enterprise C# repository.
- Engineered core modules for TQA, a Siemens NX CAD conformance checker; optimized unused-object analyzer to improve speed by 47%.

Research Assistant – HPC for NLP — Apr. 2024 – Dec. 2024
Jülich Supercomputing Centre (JSC), Forschungszentrum Jülich — *Jülich, Germany*

- Built HPC evaluation pipelines to construct a multilingual embedding proximity map across 24 languages for Teuken-7B (Germany's 1st LLM), optimizing cross-lingual transfer learning and data selection.

Projects

HackerRank LLM Orchestration Hackathon | Top 4% Worldwide [Link] — Jun. 2026

- Built a multi-modal VLM claim reviewer with deterministic guardrails for business logic, ranking 2nd in Germany.

Software Project Lab – Artifaix (AI-supported Information Foraging) [Link] — Oct. 2025 – Apr. 2026
Research Group Software Construction (SWC), RWTH Aachen — *Aachen, Germany*

- Led a 4-person team to design a modular RAG-based microservice architecture for automated knowledge extraction.
- Implemented LLM reasoning and MapReduce processing for papers exceeding model context windows; Grade 1.0.

Bachelor's Thesis – Exploring Linguistic Proximity [Link] — Jun. 2024 – Dec. 2024
Visual Computing Institute, RWTH Aachen & Forschungszentrum Jülich & WestAI — *Aachen, Germany*

- Analyzed 6.1 TB of multilingual text data from 24 languages, applied t-SNE to map high-dimensional clusters discovering that raw embeddings cluster by topic, and mean embeddings cluster by linguistic family.

Software Project Lab – Network Engineering Practical Internet eXperience (PIX) — Oct. 2022 – Apr. 2023
Chair of Communication and Distributed Systems (ComSys), RWTH Aachen — *Aachen, Germany*

- Configured OSPF, BGP, and RPKI protocols in a simulated network of 28 virtual routers.

System Programming Lab – Embedded Systems Programming — Oct. 2021 – Apr. 2022
Chair of Embedded Software, RWTH Aachen — *Aachen, Germany*

- Developed low-level C firmware for ATMega644 embedded system, managing memory and peripherals (GPIO, Timers).

Technical Skills

Programming: Python, C# (.NET), Java, C/C++, Bash, TypeScript, HTML/CSS
Machine Learning & NLP: PyTorch, Hugging Face (Transformers, Datasets), Sentence Transformers, scikit-learn, Vector Embeddings, RAG, LangChain, LangGraph, ChromaDB, MCP
Data & Process Mining: pandas, NumPy, Data Visualization, ProM, Celonis
Backend & Systems: FastAPI, Microservices, LINQ, async/await, Linux, Docker, SLURM, MapReduce, Vector Databases
Databases & Infrastructure: MySQL, SQL, Git, Conda/venv/uv
Frontend & Collaboration: React, Vite, Tailwind CSS, shadcn/ui, Jira, Confluence
Engineering Tools: LaTeX, RapidMiner, Siemens NX, NX Open API, Visual Studio

Languages

German: DSH-3 (C2 / Fluent) **English:** Fluent **Persian:** Native **Azerbaijani:** Native **Turkish:** Fluent